Geometry & Measurement

Angle facts

1 Two angles on a straight line measure $3x$ and $2x + 30$. Find x.

2 Find the angle vertically opposite a 65° angle.

3 Three angles around a point measure 110°, 95°, and x. Find x.

4 Two angles are complementary (sum to 90°). One measures 37°. Find the other.

Parallel lines and transversals

5 Two parallel lines are cut by a transversal, creating an angle of 70°. Find its corresponding angle.

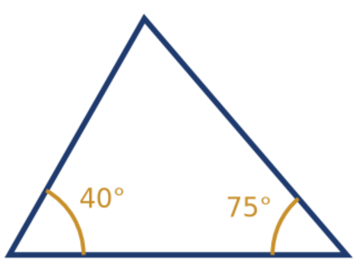
6 In the same setup, find the co-interior (allied) angle to a 70° angle.

7 Find the alternate interior angle to a 55° angle.

8 Two parallel lines cut by a transversal create an angle of $4x$ and its co-interior angle of $2x + 30$. Find x.

Triangle angle sum

9 Find the third angle of the triangle shown, given the two marked angles.



10 An isosceles triangle has two equal angles of 72° each. Find the third angle.

11 In a right triangle, one of the non-right angles is 35°. Find the other non-right angle.

12 A triangle has three equal angles. Find each angle, and name this type of triangle.

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Angle sums of polygons

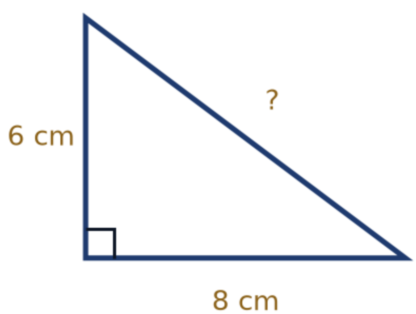
13 Find the sum of the interior angles of a hexagon (6 sides).

14 Find the sum of the interior angles of a pentagon (5 sides).

15 A regular polygon has an interior angle sum of $1{,}080^\circ$. How many sides does it have?

The Pythagorean theorem

16 Find the hypotenuse of the right triangle shown.



17 A right triangle has hypotenuse 13 and one leg 5. Find the other leg.

18 A 10 m ladder leans against a wall with its foot 6 m from the wall's base. How high up the wall does it reach?

19 A right triangle has legs 9 and 12. Find the hypotenuse.

Perimeter and area of rectangles and squares

20 A rectangle has length 12 cm and width 7 cm. Find its perimeter and area.

21 A square has side length 9 cm. Find its perimeter and area.

22 A rectangle has perimeter 50 cm and length 16 cm. Find its width and area.

23 A square has area 121 cm^2. Find its side length and perimeter.

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Area of triangles and parallelograms

24 A triangle has base 14 cm and height 9 cm. Find its area.

25 A parallelogram has base 12 cm and height 8 cm. Find its area.

26 A triangle has area 60 cm^2 and base 15 cm. Find its height.

27 A parallelogram has area 126 cm^2 and height 9 cm. Find its base.

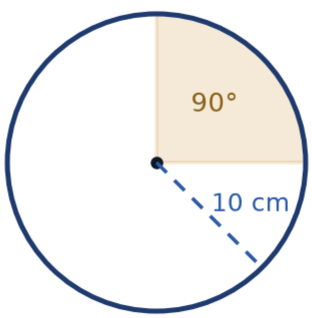
Circles: circumference and area

28 A circle has radius 7 cm. Find its circumference, using $\pi = \frac{22}{7}$.

29 A circle has radius 7 cm. Find its area, using $\pi = \frac{22}{7}$.

30 A circle has diameter 20 cm. Find its circumference in terms of π.

31 Find the area of the shaded 90° sector shown, in terms of π.



Composite figures

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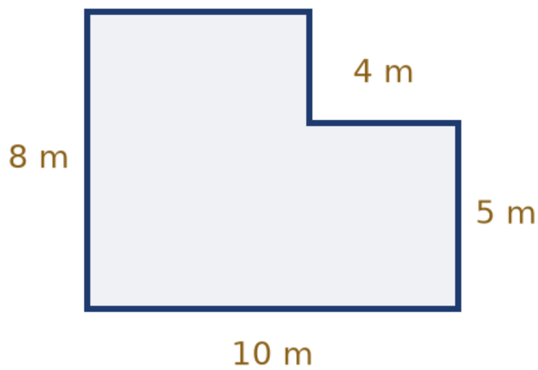
32 Find the area of the L-shaped room shown.



33 A garden is L-shaped: a 15 m by 6 m rectangle with a 5 m by 2 m rectangle cut from one corner. Find its area.

34 A shape consists of a 12 cm by 5 cm rectangle with a right triangle of base 6 cm and height 5 cm attached to one of the short sides. Find the total area.

Volume and surface area

35 A rectangular box has dimensions 5 cm $\times 4$ cm $\times 3$ cm. Find its volume.

36 A cube has edge length 6 cm. Find its volume and surface area.

37 A rectangular prism has volume 240 cm^3, length 10 cm, and width 6 cm. Find its height.

38 Find the surface area of a rectangular box with dimensions 8 cm $\times 5$ cm $\times 4$ cm.

Word problems

39 A rectangular garden measuring 20 m by 15 m is surrounded by a path 1 m wide. Find the area of the path alone.

40 New flooring costs 45 riyals per square meter. Find the total cost to floor a rectangular room measuring 6 m by 4 m.

41 A 17 m ladder leans against a vertical wall with its base 8 m from the wall. How high up the wall does it reach?

42 A jogger runs 4 laps of a circular track of radius 35 m, using $\pi = \frac{22}{7}$. Find the total distance covered.

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